57. (*d*) A number is divisible by 16 if the number formed by its last 4 digits is divisible by 16.

$\therefore$ Required remainder

$= \text{Rem.}\left(\dfrac{1920}{16}\right) = 0$

58. (*b*) Required probability $= \dfrac{{}^4C_2 \times {}^8C_1}{{}^{12}C_3}$

$= \dfrac{6\times 8}{220} = \dfrac{12}{55}$

59. (*c*) Required number of carpets

$= \dfrac{960}{6\times 4} = 40$

60. (*b*) Let the distance by *d* km.

Then, $\dfrac{d}{3} - \dfrac{d}{4} = \dfrac{15+15}{60}$

$\Rightarrow \dfrac{d}{12} = \dfrac{1}{2} \Rightarrow d = 6$

61. (*a*) Area left for writing = (30 – 2 × 2.5) × (15 – 2 × 1.5) = 25 × 12 cm^2

Required percentage

$= \dfrac{25\times 12}{30\times 15}\times 100 = 66.66\%$

62. (*b*) Let the weight of Ram and Shyam be 6 kg and 5 kg respectively.

Increased weight of Shyam = (6 + 5) × 1.15 – 6 × 1.2

= 12.65 – 7.2 = 5.45 kg

Required percentage $= \dfrac{5.45-5}{5}\times 100 = 9\%$

63. (*a*) Total quantity of water in new mixture

$= 2 + \dfrac{1}{7}\times 14 = 4$ litres

Total quantity of milk in new mixture

= (8 + 14) – 4 = 18 litres

Required ratio = 4 : 18 = 2 : 9

64. (*d*) P → CP = *x*, Profit = 0.4*x*, SP = 1.4*x*

Q → SP = 1.4*x*, Profit = 0.5 1.4 *x* = 0.7 *x*

Now, 0.7*x* – 0.4*x* = 600

$\Rightarrow 0.3x = 600 \Rightarrow x = 2000$

$\therefore$ SP = 1.4 × 2000 = 2800

65. (*b*) Let the age of the youngest child be *x* years.

According to the question,

25 × 5 + 8 × 5 + *x* + *x* + 4 = 25 × 7

$\Rightarrow x + x = 175 - 169$

$\Rightarrow 2x = 6 \Rightarrow x = 3$

66. (*a*) Distance travelled by the truck in 10 litres of diesel at the speed of 50 km/h

= 10 × 19.5 = 195 km

$\therefore$ Required distance $= \dfrac{195}{1.3} = 150$ km

67. (*a*) At 't' minutes after 2:00,

Hour hand angle = 60 + 0.5t degrees

Minute hand angle = 6t degrees

As, 1 minute space = 6 degrees,

So, 6t – (60 + 0.5t) = 6

$\Rightarrow$ 5.5t = 66 $\Rightarrow$ t = 12

Hence, at 2:12 the minute hand is exactly 1 minute space ahead of hour hand.

68. (*c*)

	C1	C2	C3
Profit ⟶	9	10	8
	81	90	72

	C2	C4	C5
Profit ⟶	18	19	20
	90	95	100

$\therefore$ Profit ratio of C1, C2, C3, C4 and C5

= 81 : 90 : 72 : 95 : 100

$\therefore$ Total profit $= \dfrac{19}{100-81} \times (81 + 90 + 72$ $+ 95 + 100) =$ ₹ 438 Cr

69. (*a*) Let the age of younger child be *x* years.

According to question,

4 × 40 + 4 × 15 + *x* + *x* + 8 = 6 × 40

$\Rightarrow 228 + 2x = 240 \Rightarrow x = \dfrac{12}{2} = 6$

Required ratio = (6 + 8) : 6 = 7 : 3

70. (*b*)

	Ramesh	Rakesh
CP ➝	x	
Profit ➝	$\dfrac{x}{4}$	$\dfrac{1}{4}\times\dfrac{5}{4}x = \dfrac{5}{16}x$
SP ➝	$x + \dfrac{x}{4} = \dfrac{5}{4}x$	$\dfrac{5}{4}x$

Now,

$\dfrac{5x}{16} - \dfrac{x}{4} = 100 \Rightarrow \dfrac{x}{16} = 100 \Rightarrow x = 1600$

SP $= \dfrac{5}{4}\times 1600 =$ ₹ 2000

71. (*c*) $\dfrac{2}{3}$ day → 18 books

1 days → 18 × $\dfrac{3}{2}$ = 27 books

$\dfrac{1}{4}$ day → $\dfrac{27}{4}$ books

72. (*b*) Upstream speed $= \dfrac{140}{7} = 20$ km/h

Downstream speed $= \dfrac{210}{7} = 30$ km/h

$\therefore$ Speed in still water

$= \dfrac{20+30}{2} = 25$ km/h

73. (*d*) Perimeter of rectangle = Circumference of circle

$= 2 \times \dfrac{22}{7} \times 21 = 132$ cm

$\therefore 2(6x + 5x) = 132 \Rightarrow 22x = 132 \Rightarrow x = 6$

Area of rectangle $= 6x \times 5x = 30x^2$

$= 30 \times 6^2 = 1080$ cm^2

74. (*c*) $\sqrt{1225} \div \sqrt[3]{343} \times 45$ % of 760

= *x* × 66.67% of 45

$\Rightarrow 35 \div 7 \times \dfrac{45}{100} \times 760 = x \times \dfrac{2}{3} \times 45$

$\Rightarrow 5 \times 342 = x \times 30 \Rightarrow x = \dfrac{342}{6} = 57$

75. (*a*)

	A		B
Time ⟶	20	:	30
	2	:	3
Efficiency ⟶	3	:	2

Required difference

$= \dfrac{3-2}{3+2}\times 4000 =$ ₹ 800

76. (*b*) 1 minute-space = 6°.

At 't' minutes past 3:00

Minute hand angle = 6t°

Hour hand angle = 90° + 0.5t° (since at 3:00 it's at 90° and moves 0.5°/min)

Minute hand ahead by 7 spaces

$\Rightarrow$ minute – hour = 7 × 6 = 42°

$\Rightarrow$ 6t – (90 + 0.5t) = 42

$\Rightarrow$ 5.5t – 90 = 42

$\Rightarrow$ 5.5t = 132

$\Rightarrow$ t = 24 minutes

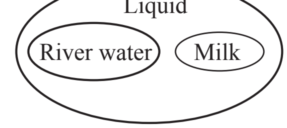
So, the time is 3:24.

77. (*a*) Logic: They're the two-letter suffixes we add to numbers when writing ordinal numbers in English.

As, 1 → st (1st), 2 → nd (2nd), 3 → rd (3rd), 4 → th (4th)

Similarly, Continuing the sequence (1st, 2nd, 3rd, 4th), the next is 5th, whose suffix is TH.

78. (*a*) Logic: "River water" and "Milk" are Liquid.

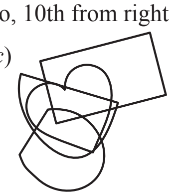


79. (*d*) After reversing each half:

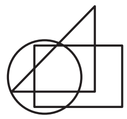
MLKJIHGFEDCBAZYXWVUTSRQPON

So, 10th from right is W.

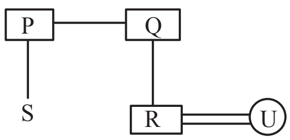
80. (*c*)



81. (*c*)



82. (*a*)



Here, U is the Sister in law of S.

83. (*b*) Logic: In Numerator each letter is decreased by 1 and denominator decreased by 3.

As,

F N B and V

$\downarrow -1$ $\downarrow -1$ $\downarrow -1$ $\downarrow -3$

E M A S

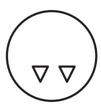
Simlarly,

G P C and Z

$\downarrow -1$ $\downarrow -1$ $\downarrow -1$ $\downarrow -3$

F O B W

84. (*a*) Logic : Second image is the water image of the first image in each pair.

? = 

85. (*b*) Political Science, History and Sociology are the parts of social sciences. But, Biology is natural science.

86. (*c*) Except number 15, all other numbers are even number.

87. (*a*) Logic : First reverse the word and then subtract 3, 2 alternatively.

As,

MOBILE in reverse order ELIBOM,

then,

E	L	I	B	O	M
$\downarrow -3$	$\downarrow -2$	$\downarrow -3$	$\downarrow -2$	$\downarrow -3$	$\downarrow -2$
B	J	F	Z	L	K

and,

TABLET in reverse order TELBAT,

then,

T	E	L	B	A	T
$\downarrow -3$	$\downarrow -2$	$\downarrow -3$	$\downarrow -2$	$\downarrow -3$	$\downarrow -2$
Q	C	I	Z	X	R

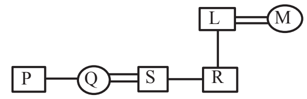
Similarly,

KINDLE in reverse order ELDNIK,

then,

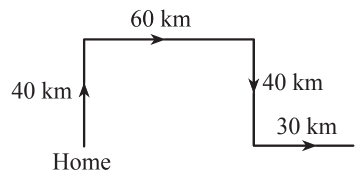
E	L	D	N	I	K
$\downarrow -3$	$\downarrow -2$	$\downarrow -3$	$\downarrow -2$	$\downarrow -3$	$\downarrow -2$
B	J	A	L	F	I

88. (*b*)



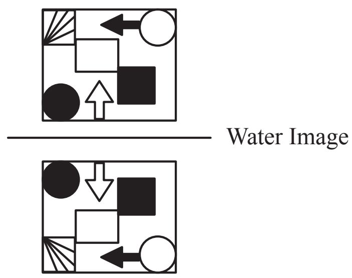
Here, P is S's wife's brother.

89. (*b*)

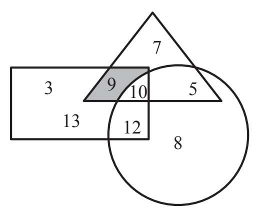


Shilpa is 90 km far from her home.

90. (*d*)



91. (*a*)



Here, 9 players play both kabaddi and football only.

92. (*d*)

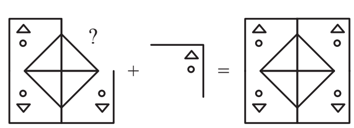
39, 53, ?, 108, 149, 199

+14 +23 +32 +41 +50

+9 +9 +9 +9

? = 53 + 23 = 76

93. (*a*)



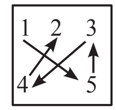
94. (*b*) Given : 105 – 15 ÷ 13 + 3 × 27 = ?

After interchanging '+' with '×; and '–' with '÷',

= 105 ÷ 15 – 13 × 3 + 27

= 7 – 39 + 27 = – 5

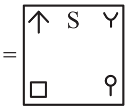
95. (*a*)



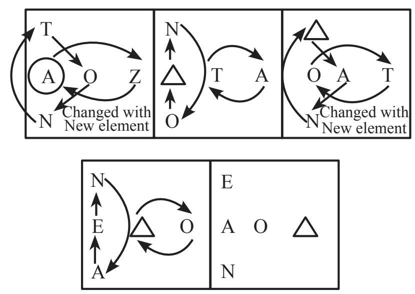
Logic :-

I. Letters or symbols changes its position in 1 → 5, 5 → 3, 3 → 4, and 4 → 2 sequence.

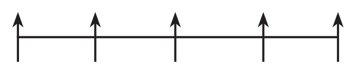
II. In the next series new symbols comes in position 1.

∴ ? = 

96. (*b*)



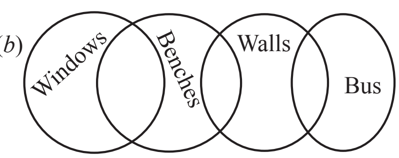
97. (*a*)



Ranjita Sucheta Amita Gargi Fauzia

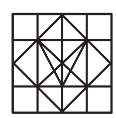
Ranjita and Fauzia sitting at the extreme ends.

98. (*b*)



Only Conclusion II follows.

99. (*b*)



100. (*c*)